

H A S E E B

VU

Virtual University

M.Sc
(MATH)

EBIT

Narowal Campus

Question #1 of 10 (Start time: 04:20:26 PM, 23 December 2021)

Total Marks: 1

If $\{s_n\}$ is -----, then $\lim_{n \rightarrow \infty} s_n = \inf \{s_n\}$.

Select the correct option

 Reload Math Equations☐

nonincreasing

☐

decreasing

☐

nondecreasing

☐

increasing

[Click to Save Answer & Move to Next Question](#)

The value of the

$$\lim_{n \rightarrow \infty} \frac{1}{n^p}$$

is

Select the correct option



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1 \text{ for } p > 1.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0 \text{ for } p > 1.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1 \text{ for all } p.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 3 \text{ for all } p.$$

The series $\sum_{n=k}^{\infty} a_n$ -----, if $(-1)^n a_n > 0$, $|a_{n+1}| < |a_n|$, and $\lim_{n \rightarrow \infty} a_n = 0$.

Select the correct option

 Reload Math Equations

diverges

converges

Click to Save Answer & Move to Next Question

$$\limsup_{n \rightarrow \infty} s_n = \infty \text{ if}$$

Select the correct option

 Reload Math Eq $\{s_n\}$ has limit inferior. $\{s_n\}$ has unique limit. $\{s_n\}$ is not bounded above. $\{s_n\}$ is convergent.

Click to Save Answer & Move to Next Qu

If $\sum a_n$ -----, then

Select the correct option

- | | |
|----------------------------------|--|
| ☒ | $\lim_{n \rightarrow \infty} a_n = 0$ |
| ☐ | $\lim_{n \rightarrow \infty} a_n = -1$ |
| ☐ | $\lim_{n \rightarrow \infty} a_n = 1$ |
| ☐ | None of these |

Click to

The limit of the sequence

$$s_n = \frac{1}{n} + \frac{2(1 + 3/n)}{1 + 1/n}.$$

Select the correct option



2



3



1



4

A point x is a limit point of a set iff there is a sequence $\{x_n\}$ of point in S such that $x_n \neq x$ for $n \geq 1$, and - - - - -

Select the correct option

- | | |
|----------------------------------|--|
| ☒ | $\lim_{n \rightarrow \infty} x_n = x$ |
| ☐ | $\lim_{n \rightarrow \infty} x_n > x$ |
| ☐ | $\lim_{n \rightarrow \infty} x_n \neq x$ |
| ☐ | None of these |

Click to Save Answer & Move

If $\{x_n\}$ is ---, then $\{x_n\}$ has a convergent subsequence.

Select the correct option



unbounded

bounded



Click to Save Answer & Mark

Question # 1 of 10 (Start time: 04:27:17 PM, 23 December 2021)

$$\liminf_{n \rightarrow \infty} s_n = -\infty \text{ if}$$

Select the correct option

☐ $\{s_n\}$ is not bounded above.☐

$$\lim_{n \rightarrow \infty} s_n = 0.$$

☐ $\{s_n\}$ is convergent.☐ $\{s_n\}$ is not bounded below.

Click to Save Answer

Question # 2 of 10 (Start time: 04:27:55 PM, 23 December 2021)

The given series $\sum_{n=1}^{\infty} \frac{1}{n}$ is -----.

Select the correct option

convergent



divergent



Time: 04:29:07 PM, 23 December 2021)

If $\{s_n\}$ is a cauchy sequence of real numbers, then $\{s_n\}$ is — — — —.

unbounded

bounded



Click to Save Answer & Move to Next Question

Question # 5 of 10 (start time: 04:29:39 PM, 23 December 2021)

If $\sum a_n$ -----, then

Select the correct option

 Reload Mo☐

None of these

☐ $\lim_{n \rightarrow \infty} a_n = 0$ ☐ $\lim_{n \rightarrow \infty} a_n = 1$ ☐ $\lim_{n \rightarrow \infty} a_n = -1$ [Click to Save Answer & Move to the](#)

of 10 (start time: 04:30:12 PM, 23 December 2021)

If $\{s_n\}$ is ———, then $\lim_{n \rightarrow \infty} s_n = \inf \{s_n\}$.

Correct option



Reload M

nonincreasing

nondecreasing

decreasing

increasing

Click to Save Answer & Move to N

Question # 7 of 10 (Start time: 04:30:56 PM, 23 December 2021)

If $\{s_n\}$ is bounded above and does not diverge to $-\infty$, then there is a unique real number $\bar{s}$ such that
if $\varepsilon > 0$, $s_n > \bar{s} - \varepsilon$ — — — — —

Select the correct option

 Reloadfor infinitely many n for finitely n

Click to Save Answer & Move to

Question # 8 of 10 (Start time: 04:31:46 PM, 23 December 2021)

Total Marks: 1

If $\{s_n\}$ is a sequence of real numbers, then $\lim_{n \rightarrow \infty} s_n = s$ iff $\lim_{n \rightarrow \infty} \sup s_n = \dots = \lim_{n \rightarrow \infty} \inf s_n = s$.

Select the correct option

Reload Math Equations

 $=$  $<$  $>$ 

None of these

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 04:32:28 PM, 23 December 2021)

If the sequence is convergent then

Select the correct option

☐

it is bounded above but may not be bounded below.

☐

it has two limits.

☒

it is bounded.

☐

it is bounded below but may not be bounded above.

Question # 1 of 10 (Start time: 04:34:04 PM, 23 December 2021)

The value of the $\lim_{n \rightarrow \infty} \frac{1}{n^p}$ is

Select the correct option

☐ $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 3$ for all p .

☐ $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1$ for all $p > 1$.

☒ $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$ for $p > 0$.

☐ $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1$ for all p .

A sequence $\{s_n\}$ of real numbers is called a Cauchy sequence if for every $\epsilon > 0$, there is an integer N such that $|s_n - s_m| < \epsilon$ if $m, n \geq N$.

Select the correct option



Reload Math Equations



>



$\leq$



$\geq$



<



Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 04:35:16 PM, 23 December 2021)

For the convergent sequence

$$\{s_n\},$$

which statement is true?

Select the correct option

☐

$$\limsup_{n \rightarrow \infty} s_n \neq \liminf_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} s_n.$$

☐

has a divergent subsequence.

☒

$$\limsup_{n \rightarrow \infty} s_n = \liminf_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} s_n.$$

☐

$$\limsup_{n \rightarrow \infty} s_n \neq \liminf_{n \rightarrow \infty} s_n \neq \lim_{n \rightarrow \infty} s_n.$$

Question # 4 of 10 (Start time: 04:35:54 PM, 23 December 2021)

The series $\sum_{n=0}^{\infty} r^n$, $-1 < r < 1$, is

Select the correct option



divergent.



convergent and converges to $\frac{1}{1-r}$.



convergent and converges to $\frac{1}{1+r^2}$.



convergent and converges to 5.



If $\sum a_n$ ----, then

Select the correct option

☐

None of these

☐

$$\lim_{n \rightarrow \infty} a_n = 0$$

☐

$$\lim_{n \rightarrow \infty} a_n = -1$$

☐

$$\lim_{n \rightarrow \infty} a_n = 1$$



If a sequence $\{s_n\}$ is nonincreasing, then — — — —

Select the correct option

 Reload



$$\lim_{n \rightarrow \infty} s_n = \infty$$



$$\lim_{n \rightarrow \infty} s_n = \sup \{s_n\}$$



None of these



$$\lim_{n \rightarrow \infty} s_n = \inf \{s_n\}$$



If $\{s_n\}$ is a cauchy sequence of real numbers, then $\{s_n\}$ is - - - - -.

Select the correct option

☐	<i>bounded</i>
☐	<i>unbounded</i>



Reload

Click to Save Answer & Move to

Question # 6 of 10 (Start time: 04:45:28 PM, 23 December 2021)

The value of the $\lim_{n \rightarrow \infty} \frac{1}{n^p}$ is

Select the correct option



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1 \text{ for all } p > 1.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1 \text{ for all } p.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0 \text{ for } p > 0.$$



$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 3 \text{ for all } p.$$

Question # 7 of 10 (start time: 04:46:07 PM, 23 December 2021)

If a sequence $\{s_n\}$ is nonincreasing, then - - - -

Select the correct option



Reload M



$$\lim_{n \rightarrow \infty} s_n = \sup \{s_n\}$$



None of these



$$\lim_{n \rightarrow \infty} s_n = \infty$$



$$\lim_{n \rightarrow \infty} s_n = \inf \{s_n\}$$

Click to Save Answer & Move to N

Question # 4 of 10 (Start time: 04:53:12 PM, 23 December 2021)

Total Marks: 1

If $\{a_n\}_k^\infty$ is an infinite sequence of real numbers, the symbol $\sum_{n=k}^\infty a_n$ is ----- and a_n is the n^{th} term of series.

Select the correct option

 Reload Math Equations☐ finite series☒ infinite series
Click to Save Answer & Move to Next Question

Question # 5 of 10 (**Start time: 04:54:11 PM, 23 December 2021**)

The value of the $\lim_{n \rightarrow \infty} \frac{1}{n^p}$ is

Select the correct option

☐	$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1$ for all $p > 1$.
☐	$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$ for $p > 0$. 
☐	$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 3$ for all p .
☐	$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 1$ for all p .

Question # 6 of 10 (Start time: 04:54:48 PM, 23 December 2021)

Let $\sum_{n=k}^{\infty} a_n = A$ and $\sum_{n=k}^{\infty} b_n = B$, where A and B are finite. Then $\sum_{n=k}^{\infty} ca_n = \dots$

Select the correct option

- | | |
|----------------------------------|---------------|
| ☒ | cA |
| ☐ | $c - A$ |
| ☐ | $c + A$ |
| ☐ | $\frac{c}{A}$ |

Click to Save Answer

36 PM, 23 December 2021)

If $\{s_n\}$ is ———, then $\lim_{n \rightarrow \infty} s_n = \inf \{s_n\}$.

decreasing

nondecreasing

increasing

nonincreasing



Question # 8 of 10 (Start time: 04:56:19 PM, 23 December 2021)

Which statement(s) is(are) true for the following sequence $\{\sqrt[n]{n}\}$

Select the correct option

☐

The sequence is convergent and converges to 1.

☐

The sequence is divergent to ∞ .

☐

The sequence is divergent.

☐

The sequence is convergent and converges to 0.

MTH621 – Real Analysis I (Quiz-1)

Question # 9 of 10 (**Start time: 04:57:19 PM, 23 December 2021**)

$$\limsup_{n \rightarrow \infty} s_n = \infty \text{ if}$$

Select the correct option

☐	$\{s_n\}$ has limit inferior.
☐	$\{s_n\}$ is convergent.
☐	$\{s_n\}$ has unique limit.
☐	$\{s_n\}$ is not bounded above.

Filters



If $\sum a_n$ -----, then $\lim_{n \rightarrow \infty} a_n = 0$

Select the correct option



Reload Math Equations

☒ converges

☐ diverges

Click to Save Answer & Move to Next Question

A sequence $\{s_n\}$ of real numbers is called a Cauchy sequence if for every $\varepsilon > 0$, there is an integer N such that $|s_n - s_m| < \varepsilon$ if $m, n \geq N$.

Select the correct option

 Reload Math Equations

☐	$\leq$
☒	$\geq$
☐	$>$
☐	$<$



Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 07:12:45 PM, 23 December 2021)

Total Marks: 1

If $\{x_n\}$ unbounded below, then $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that $\lim_{k \rightarrow \infty} x_{n_k} = \text{-----}$

Select the correct option



Reload Math Equations

☐	0
☐	∞
☐	1
☒	$-\infty$

Click to Save Answer & Move to Next Question

$$\limsup_{n \rightarrow \infty} s_n = \infty \text{ if}$$

Select the correct option

☐ $\{s_n\}$ has limit inferior.

☐ $\{s_n\}$ is convergent.

☐ $\{s_n\}$ has unique limit.

☒ $\{s_n\}$ is not bounded above.



Which statement(s) is(are) true about the following sequence $s_0 = 1$ and $s_n = 1 - e^{-s_{n-1}}$

Select the correct option



$$s_{n+1} - s_n = -(e^{-s_n} - e^{-s_{n-1}})$$



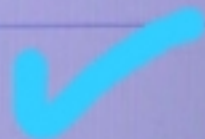
All of the above.



$$0 < s_n \leq 1 \text{ for all } n.$$



Sequence is convergent.



Question #1 of 10 (Start time: 06:38:03 PM, 23 December 2021)

Total Marks: 1

The set of terms of $\{s_{n_k}\}$ is contained in the set of terms of $\{s_n\}$ implies

Select the correct option

 Reload Math Equations

☐ $\sup\{s_n\} = \sup\{s_{n_k}\}.$

☐ $\sup\{s_n\} \leq \sup\{s_{n_k}\}.$

☒ $\sup\{s_n\} \geq \sup\{s_{n_k}\}.$

☐ $\{s_n\}$ is convergent.

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 06:35:57 PM, 23 December 2021)

Total Marks: 1

A divergent infinite series that does not diverge to $-\infty$ or $+\infty$ is said to be oscillate

Select the correct option

 Reload Math Equations☐ ± 1 ☐ ± 2 ☐

None of these

☒ $\pm \infty$  Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 06:35:29 PM, 23 December 2021)

Total Marks: 1

If $\{a_n\}_k^\infty$ is an infinite sequence of real numbers, the symbol $-\ - - - -$ is an infinite series and is the n^{th} term of series.

Select the correct option

 Reload Math Equations

None of these



$$\prod_k^\infty a_n$$



$$\sum_{n=k}^\infty a_n$$



$$a_n$$

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 06:34:48 PM, 23 December 2021)

Total Marks: 1

For harmonic series $\sum_{n=1}^{\infty} \frac{1}{n} = \text{-----}$,

Select the correct option

 Reload Math Equations

1



0



-1



None of these



Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 06:34:18 PM, 23 December 2021)

Total Marks: 1

If $f(x) = \sqrt{4-x^2}$ and $g(x) = \sqrt{x-1}$, then domain of the quotient function f/g is

Select the correct option

 Reload Math Equations $[-3, 3]$  $[1, 2]$  $[7, 8]$  $(1, 2]$ 

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 06:28:51 PM, 23 December 2021)

Total Marks: 1

If $\{x_n\}$ unbounded above, then $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that $\lim_{k \rightarrow \infty} x_{n_k} = \dots$

Select the correct option



Reload Math Equations

 $-\infty$ 

0



1

 ∞ 

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 06:27:50 PM, 23 December 2021)

Total Marks: 1

If $|x| < 1$, then

 Reload Math Equations

Select the correct option

☐

$$\lim_{n \rightarrow \infty} x^n = 0$$

☐

None of these

☐

$$\lim_{n \rightarrow \infty} x^n = 1$$

☐

$$\lim_{n \rightarrow \infty} x^n = -1$$

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 06:25:07 PM, 23 December 2021)

Total Marks: 1

Every sequence $\{s_n\}$ of real numbers has a unique limit superior, s , and a unique limit inferior, s , in the extended reals, and

Select the correct option

 Reload Math Equations


$$s \geq S$$



$$s = S$$



None of these



$$s \leq S$$

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 06:24:26 PM, 23 December 2021)

Total Marks: 1

If $f(x) = \sqrt{4-x^2}$ and $g(x) = \sqrt{x-1}$, then domain of the quotient function f/g is

Select the correct option



Reload Math Equations

☐ $[-3, 3]$ ☐ $[1, 2]$ ☐ $(1, 2]$ ☐ $[7, 8]$

Click to Save Answer & Move to Next Question

Question #10 of 10 (Start time: 06:13:33 PM, 23 December 2021)

Total Marks: 1

If the sequence is convergent then

Select the correct option

☐

it is bounded below but may not be bounded above.

☐

it has two limits.

☐

it is bounded above but may not be bounded below.

☐

it is bounded.

[Click to Save Answer & Move to Next Question](#)

Question # 8 of 10 (Start time: 06:12:11 PM, 23 December 2021)

Total Marks:

Let $\sum_{n=k}^{\infty} a_n = A$ and $\sum_{n=k}^{\infty} b_n = B$, where A and B are finite.

Then $\sum_{n=k}^{\infty} (a_n + b_n) = \text{-----}$

Select the correct option

 Reload Math Equations☒

$A + B$

☐

$\frac{A}{B}$

☐

$A - B$

☐

AB

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 06:31:22 PM, 23 December 2021)

Total Marks: 1

The sum of a convergent series is - - - - -.

Select the correct option

 Reload Math Equations☐ different☒ unique
Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 06:08:08 PM, 23 December 2021)

Total Marks: 1

If $\{a_n\}_k^\infty$ is an infinite sequence of real numbers, the symbol $\sum_{n=k}^\infty a_n$ is an infinite series and is the n^{th} term of series.

Select the correct option



Reload Math Equations

☐

None of these

☐

$$\sum_{n=k}^{\infty} a_n$$

☐ a_n ☐

$$\prod_{n=k}^{\infty} a_n$$

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 06:01:50 PM, 23 December 2021)

Total Marks: 1

A divergent infinite series that does not diverge to $\pm \infty$ is said to be - - - -.

Select the correct option

 Reload Math Equations*convergent**oscillate**bounded**None of these*

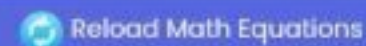
Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 05:59:28 PM, 23 December 2021)

Total Marks: 1

Which statement(s) is(are) true about the following sequence $s_0 = 1$ and $s_n = 1 - e^{-s_{n-1}}$

Select the correct option

 Reload Math Equations $0 < s_n \leq 1$ for all n . $s_{n+1} - s_n = -(e^{-s_n} - e^{-s_{n-1}})$.

Sequence is convergent.



All of the above.



Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 05:59:28 PM, 23 December 2021)

Total Marks: 1

Which statement(s) is(are) true about the following sequence $s_0 = 1$ and $s_n = 1 - e^{-s_{n-1}}$  Reload Math Equations $0 < s_n \leq 1$ for all n . $s_{n+1} - s_n = -(e^{-s_n} - e^{-s_{n-1}})$.

Sequence is convergent.



All of the above.

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 05:58:16 PM, 23 December 2021)

Total Marks: 1

Every Cauchy sequence has a

Select the correct option

☐

decreasing subsequence.

☐

convergent subsequence.

☐

positive subsequence.

☐

increasing subsequence.



Click to Save Answer & Move to Next Question

If $\{s_n\}$ is bounded above and does not diverge to $-\infty$, then there is a unique real number $\bar{s}$ such that
if $\varepsilon > 0$, $s_n < \bar{s} + \varepsilon$ -----

Select the correct option



Reload Math Equations

☐ for large n

☐ for small n

Question # 5 of 10 (Start time: 05:43:22 PM, 23 December 2021)

Total Ma

If $\{x_n\}$ unbounded below, then $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that $\lim_{k \rightarrow \infty} x_{n_k} = \text{---}$

Select the correct option

🔄 Reload Math Equation

☐ 1

☒ $-\infty$

☐ ∞

☐ 0

If $\{x_n\}$ unbounded above, then $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that $\lim_{k \rightarrow \infty} x_{n_k} = \text{---}$

Select the correct option



Reload Math Equ



∞



0



1

Question # 3 of 10 (Start time: 05:41:38 PM, 23 December 2021)

If $\sum a_n$ -----, then $\lim_{n \rightarrow \infty} a_n = 0$

Select the correct option

☐ diverges

☒ converges

Question # 2 of 10 (Start time: 05:40:42 PM, 23 December 2021)

A sequence $\{s_n\}$ of real numbers is called a ----- if for every $\varepsilon > 0$, there is an integer N such that $|s_n - s_m|$

Select the correct option

Reload



cauchy sequence



bounded sequence



None of these



sequence